# Exercise 1 : Electrical Circuits

On April 30th we this exercise together with my teammate Mai Linh

Task 1 LED Control Circuit

Task 1.1 Simple LED Circuit

the first task was to build a simple circuit with the given materials at lab to just light up an LED .

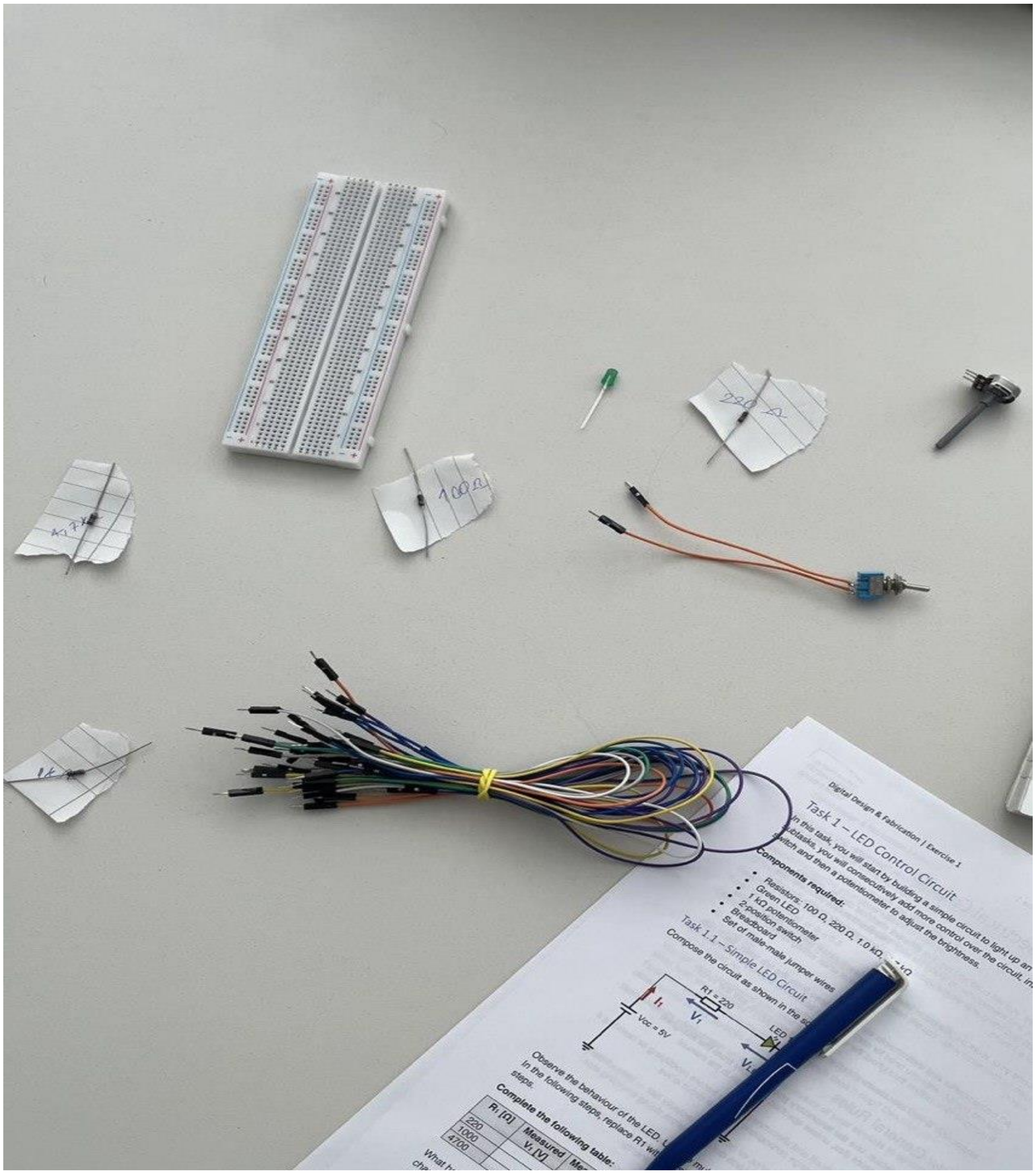
the needed components were :

A green LED

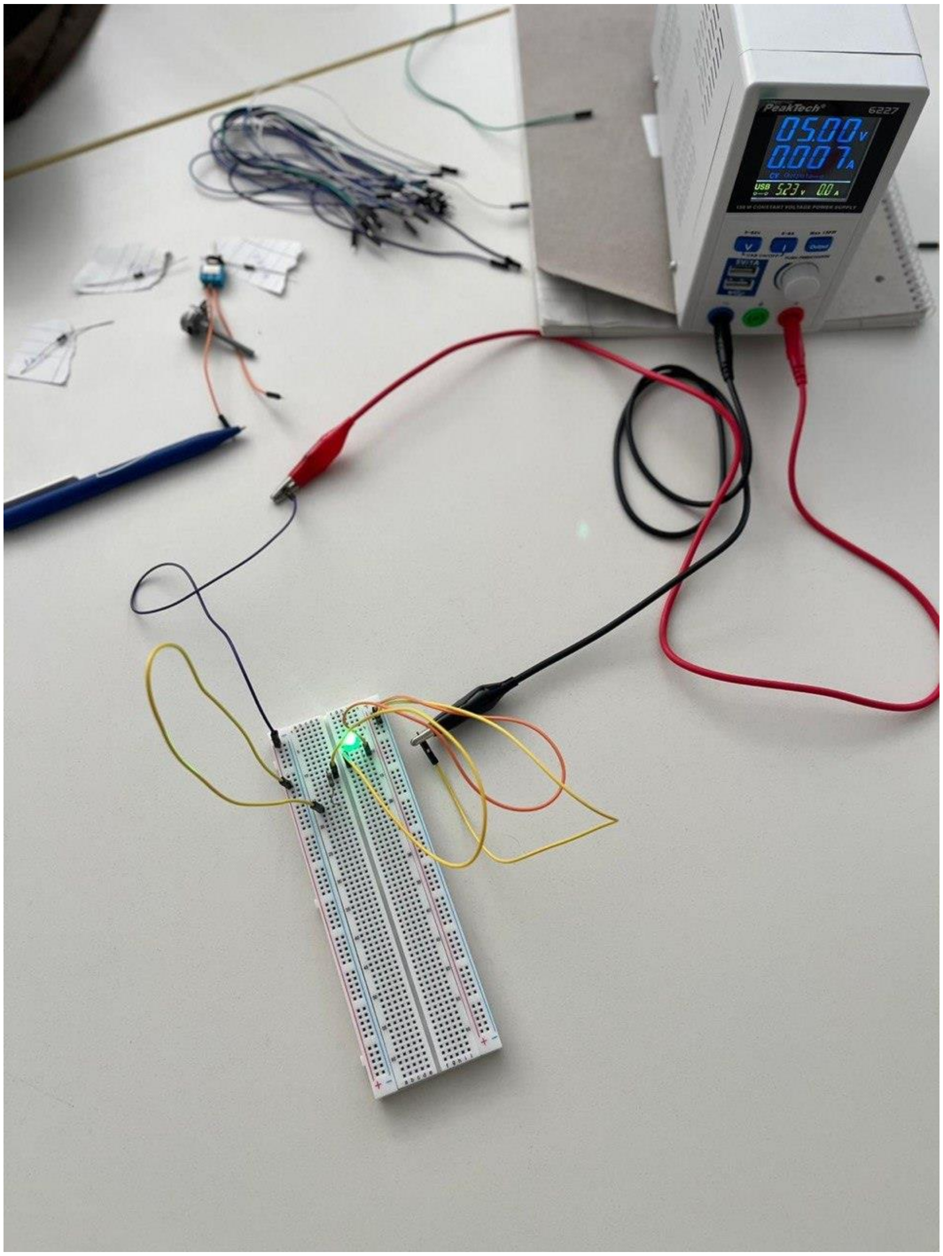
A breadboard

few male male jumper wires

Resistors: 1000 , 220 , 4700 ohm




At first we had some issue with putting the resistors in the right position and we slowly got the hang of it and it worked as in the picture :



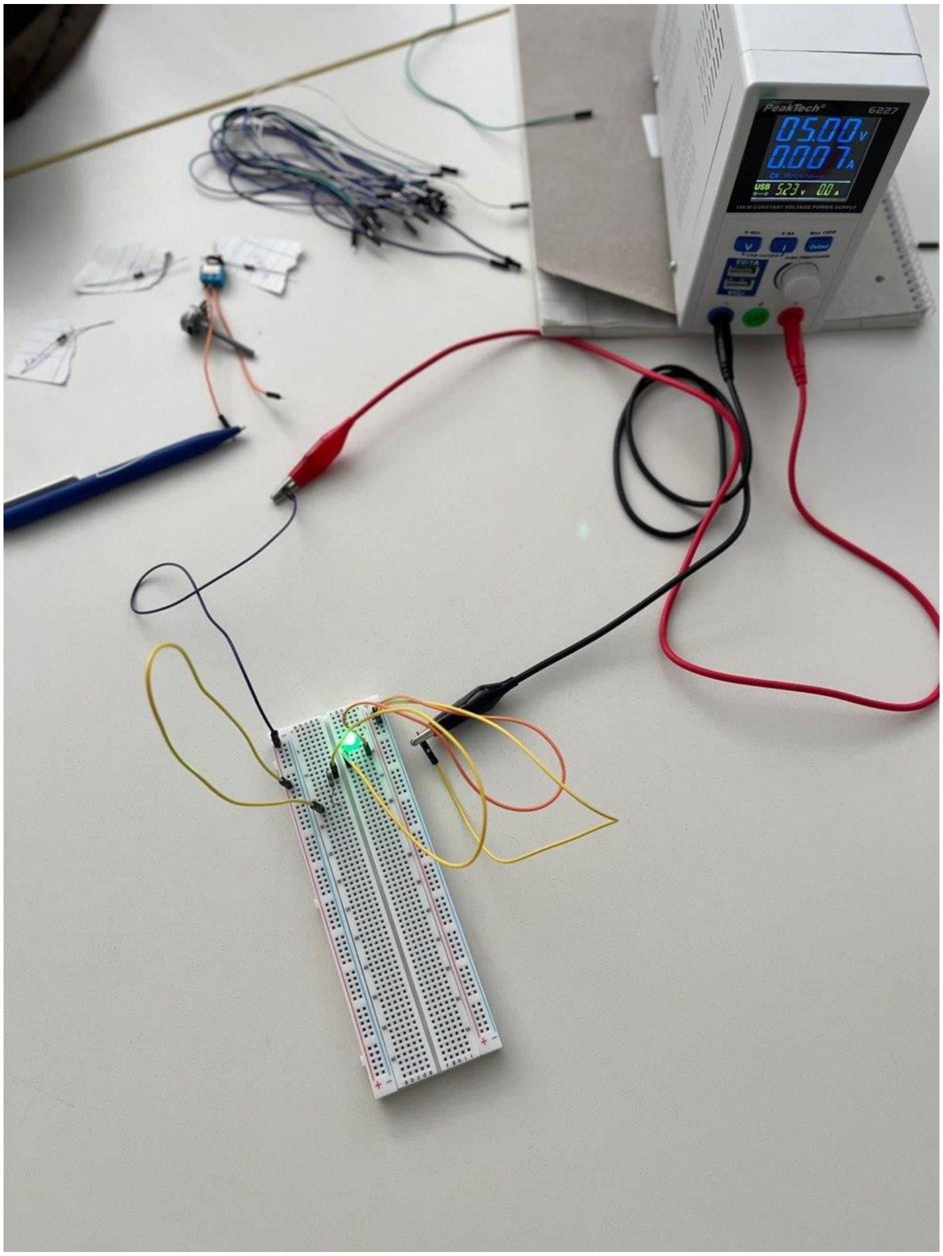

afterwards we measured Voltage and also Voltage of the LED for all three resistors.

we used multimeter and placed rods on each leg of resistor and measured .

( same process for LED as well )

at first we could not get a stable number but after placing the rods more accurately

we got the correct measurement number .




the final measurement were :

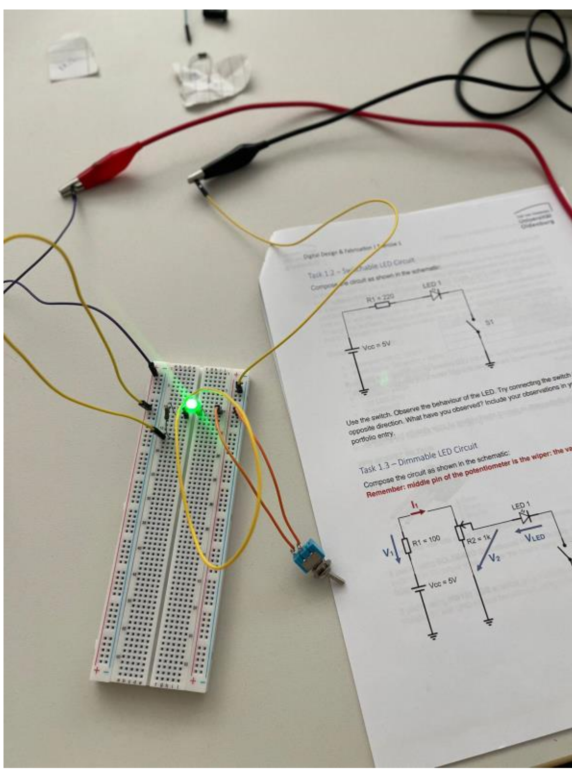
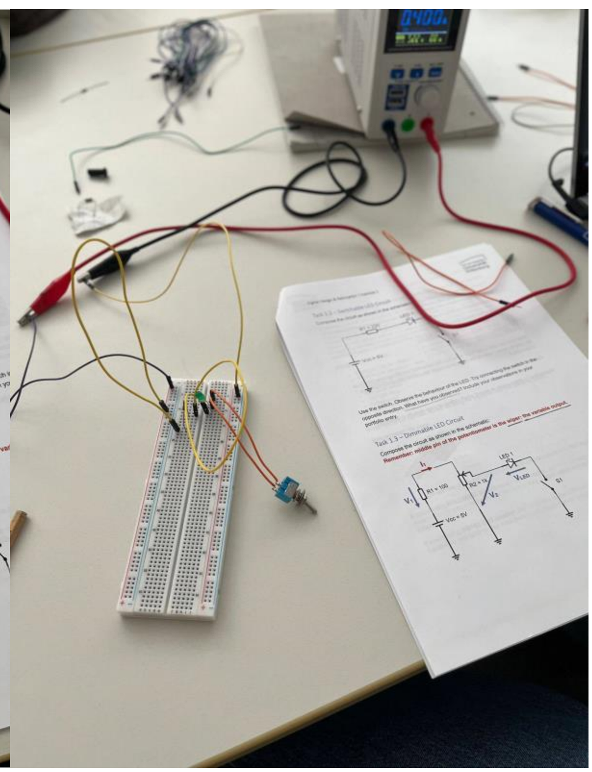
| R1 | V1 | V(LED) |
|---|---|---|
| 220 | 2.1 | 2.7 |
| 1000 | 2.5 | 2.4 |
| 4700 | 2.6 | 2.4 |

based on our measurements we got to the conclusion that as the higher the ohm got Voltage increased and V(LED) decreased , other noticeable was that as the resistant number gets higher the the LED light will get dimmer .

Task 1.2 : switchable LED  circuit

we add a switch to our circuit  and the important thing is to consider that the switch might not be in the parallel path and after we removed a cable we could easily turn the LED on and off .

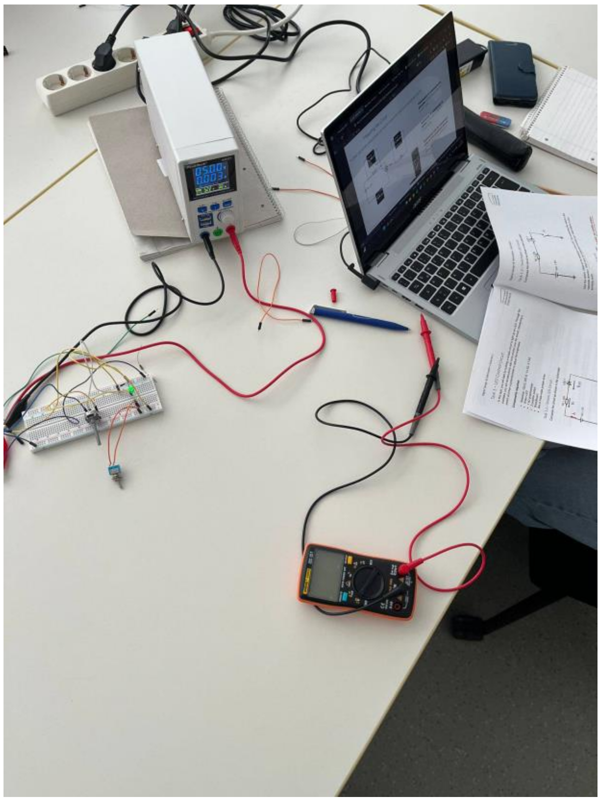
One thing that we realized was that weather we put the switch in that position or rotate it it will still work .

Task 1.3 : Dimmable LED Circuit

In this task we added the potentiometer to the circuit and after some debate together we understood that the middle pin should be connected to the LED and as for the other two pin as u can see in the image below , one goes to GND and the other to VCC .

As we switch the potentiometer on we have the LED at full brightness and when turning it slowly the LED light will decrease eventually .



(video 1)

We measured the Voltage of LED and V2 at three different state as shown in the table :

| Position | V (LED) | V 2 |
|---|---|---|
| Full brightness | 2.4 | 3.0 |
| dimmed | 2.2 | 2.1 |
| off | 0.04 | 0.08 |

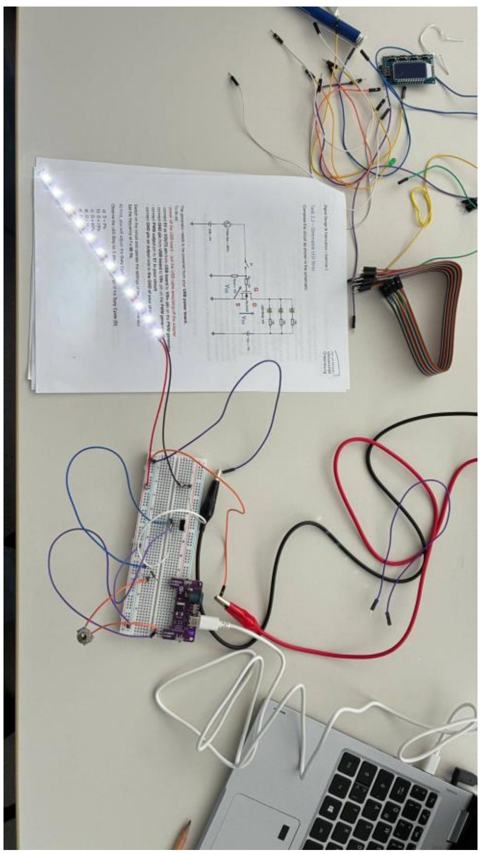
Base on the measurements we realized that we as we turn the potentiometer to the off side LED voltage will decrease and the same thing goes for V2 .

## Task 2 Transistor Switch Circuit

Task 2.1 : Switchable LED strip

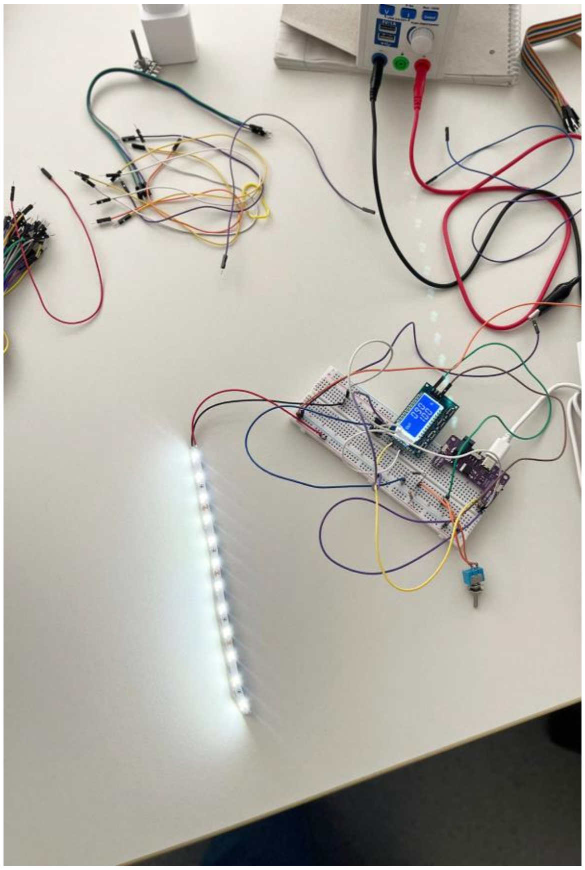
This task was about turning on LED strips and for that we used NPN MOSFET IRLZ44N and the switch was also included .

The use of the switch was to control the 5v so it won’t control the LED Voltage directly .

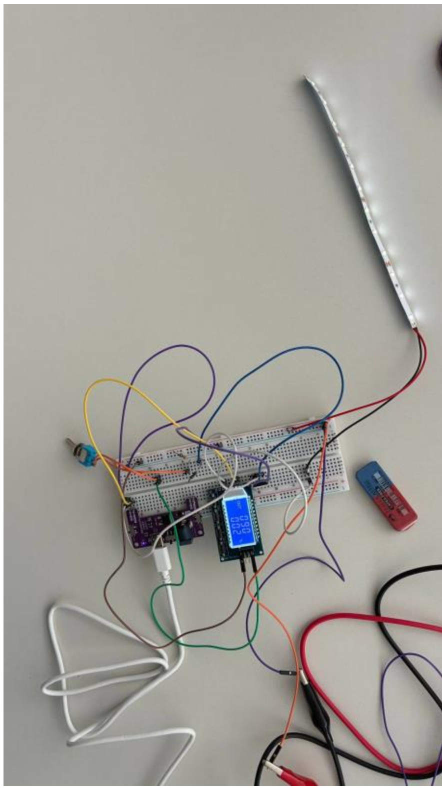
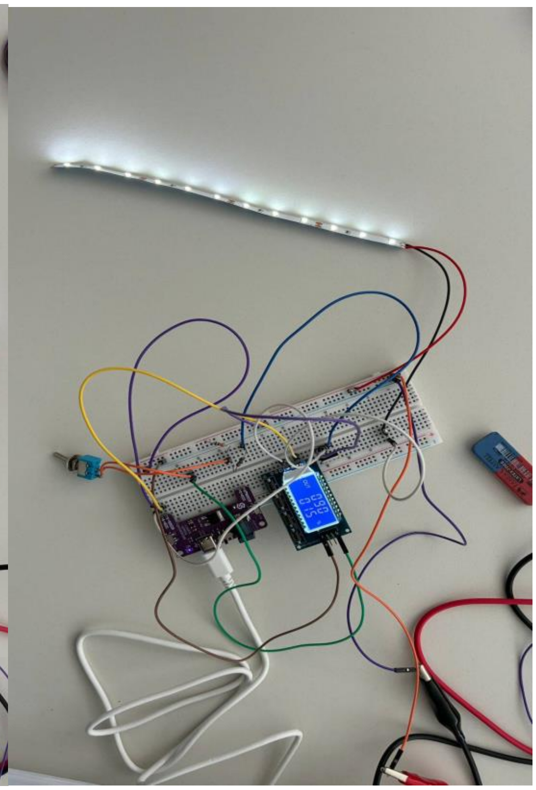
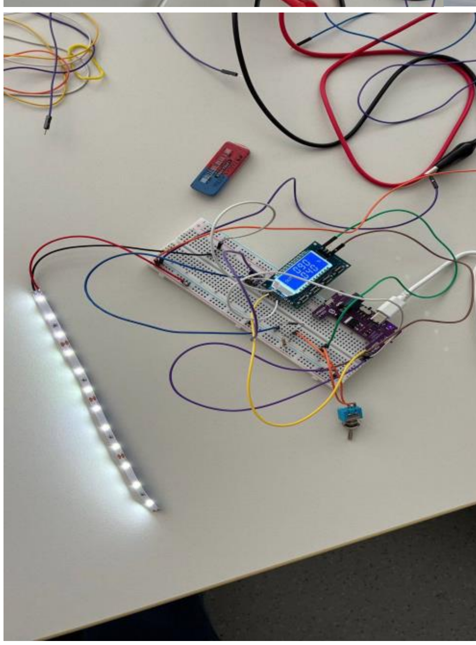
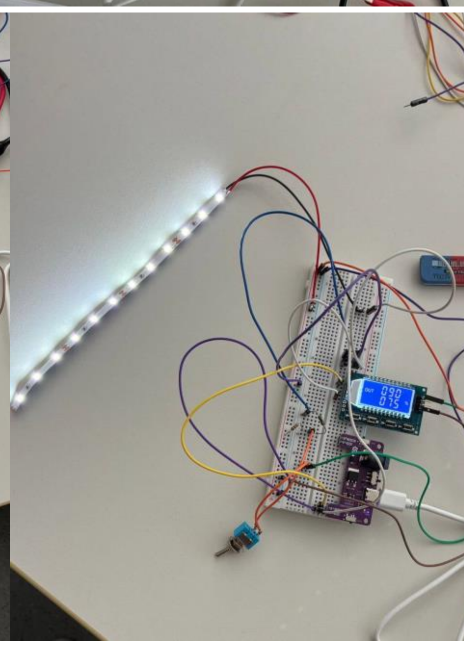


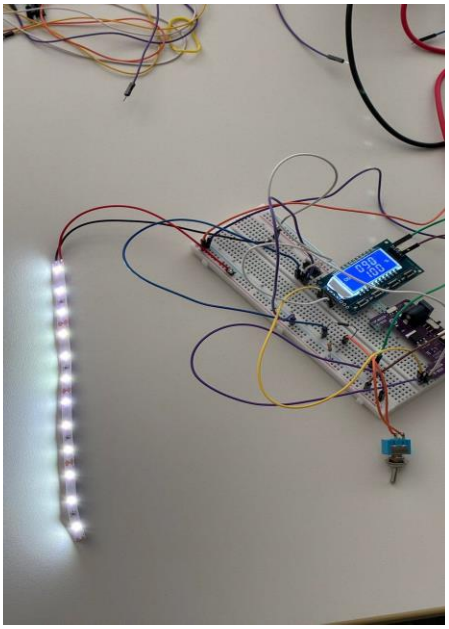
( video 2 )

Task 2.2 : Dimmable LED Strip



We added PWM generator  to circuit and based on the instruction we put the frequency to 90 Hz  and observed LED at 5 different Duty cycle stage we realized that at 100% LED had full brightness and 2% was the dimmest .



For the second part we put the duty cycle at 50%  and change the frequency we both observed that at the lower frequency the LED light  will blink and flicker but when we increase the frequency the light seems to be stable though that’s not the case , this is only because we can not really see that fast rapid change .

(video 3)